Mahmoud Alyosify

AI Instructor & Technical Trainer

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SUMMARY

AI instructor and technical trainer with **six years of teaching**: 10,000+ learners and 700K+ views of Arabic computer-science courses, and machine learning bootcamps at NTI since 2025. A practitioner who still builds: MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30), and AI Engineer Intern at Injaz Tech. Explains hard topics in plain words and grounds every lesson in hands-on labs.

SKILLS

Teaching: curriculum design, hands-on labs, exams and assessment, capstone supervision, online course production

Subjects: machine learning, deep learning, LLMs and RAG, Python, algorithms, discrete mathematics, probability and statistics

Tools: Python, scikit-learn, PyTorch, TensorFlow, LangChain, Jupyter

Delivery: Arabic and English, in person and online, recorded and live

TEACHING EXPERIENCE

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.
- Teach the **AI for Business** module of the same program, on a separate engagement.
- Write the final exams and assess the trainees' end-to-end capstone projects.

Online Instructor — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.
- Wrote, recorded and edited each course end to end, and answered learners' questions after release.

Student Ambassador — 2022 – 2023
Information Technology Institute (ITI) — Assiut

- Represented ITI on campus and guided students to its free national technical-training programs.

OTHER EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Building an LLM assistant (LangChain, LangGraph, FastAPI) and presenting each milestone in weekly study guides and demos.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

AI in Cybersecurity Intern — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.

PROJECTS

MinimalLM — Master's thesis, published on arXiv — 2026
Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).

SimCLR-Vision-SSL — self-supervised image model 2026
PyTorch, SimCLR, SupCon, ONNX, FAISS

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).

RL-Cloud-Autoscaler — deep RL for cloud scaling 2026
Python, Gymnasium, PPO, DQN

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.

Horus-OSINT — fine-tuned LLM on AWS 2026
Python, Llama-3-8B, QLoRA, PySpark, AWS (EC2, S3, EMR), Terraform

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

CERTIFICATIONS

- IBM AI Developer Professional Certificate (IBM, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)
- AWS Certified Cloud Practitioner (Amazon Web Services)
- AI in Cybersecurity (NTI, 420 hours, 2025)

AWARDS & LEADERSHIP

- 1st Place, Science & Technology Content Competition (university level)
- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Ideal Student Award, Faculty of Computers & Information, Assiut University (2022–2023)
- Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University

LANGUAGES

Arabic (native) · English (professional working proficiency)